nature energy

SUPPLEMENTARY INFORMATION

https://doi.org/10.1038/s41560-017-0032-9

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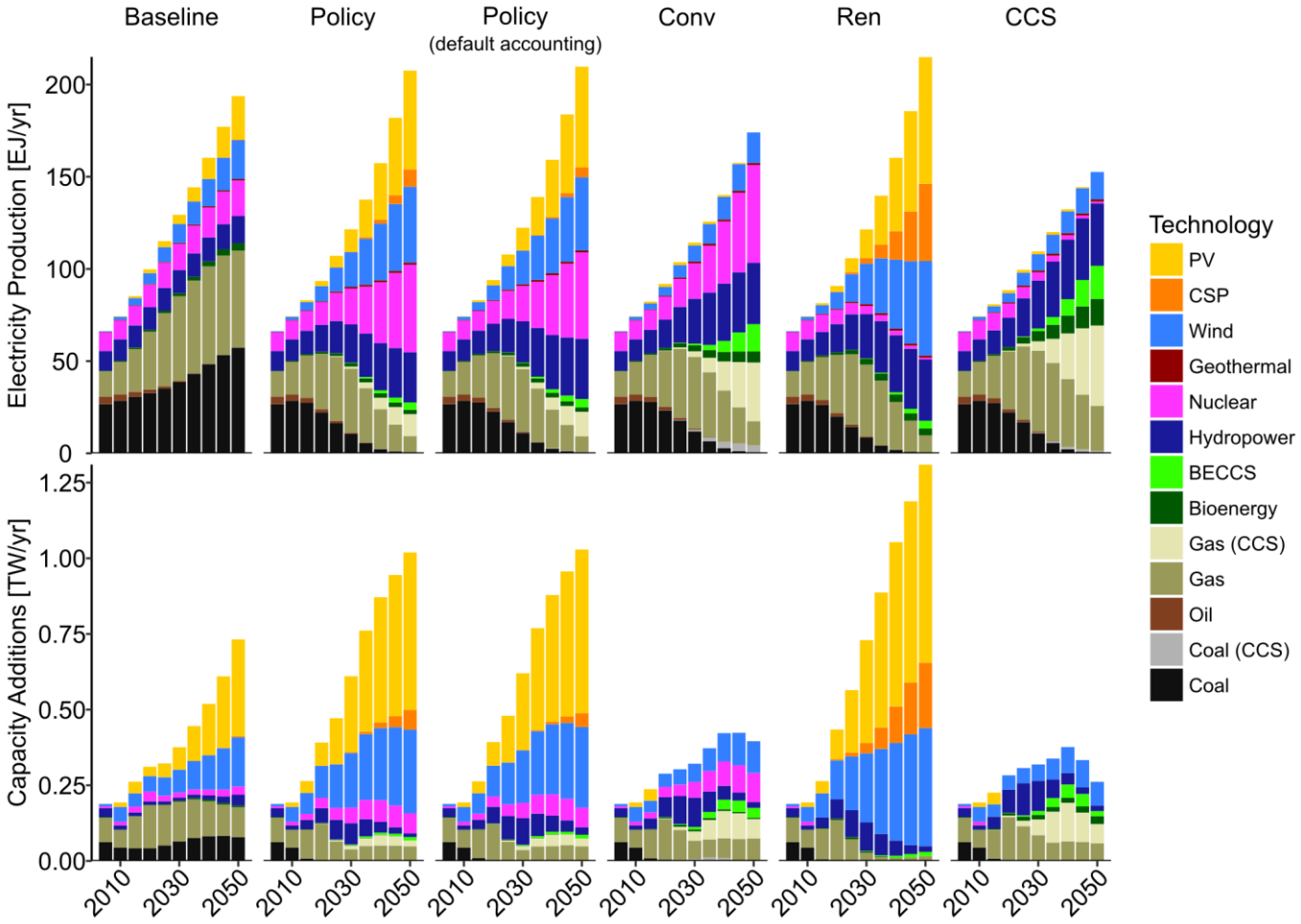
Understanding future emissions from low-carbon power systems by integration of life-cycle assessment and integrated energy modelling

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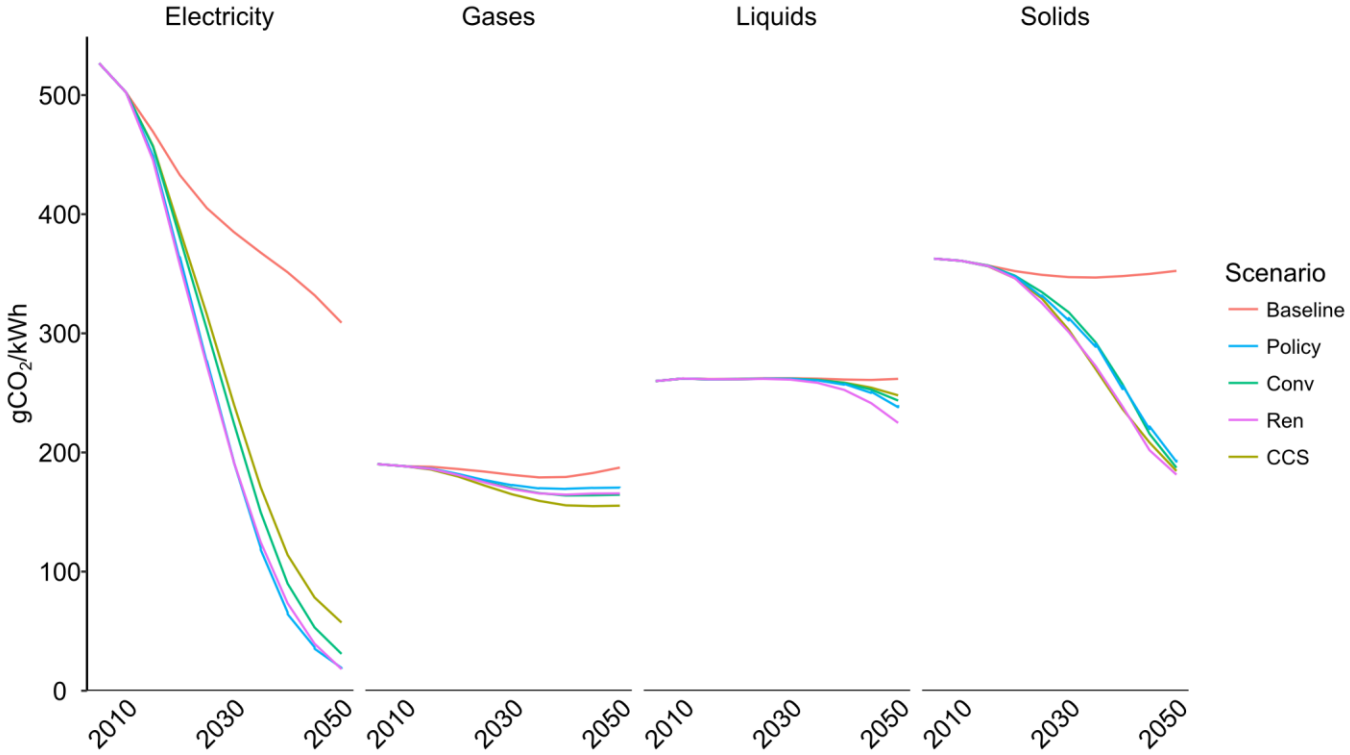
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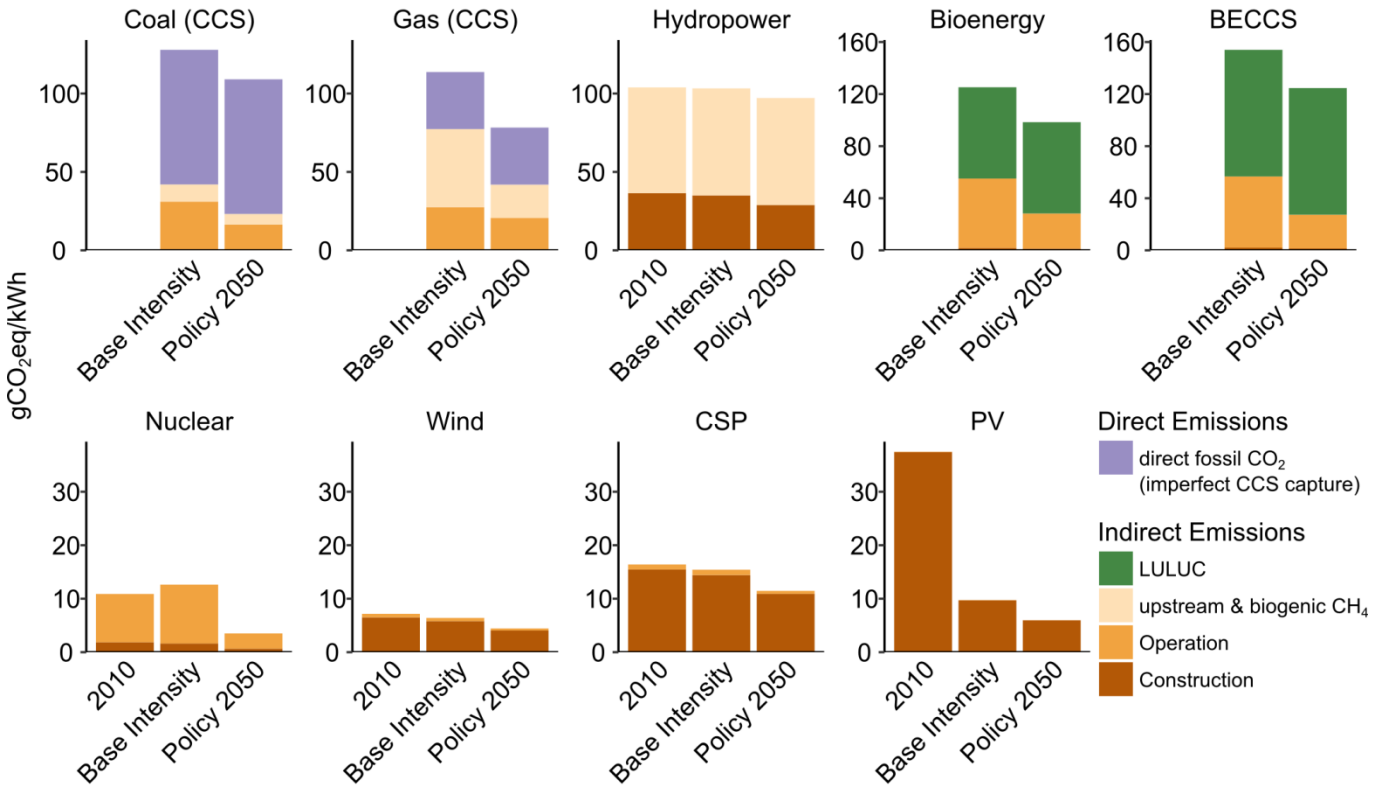
Supplementary Figures



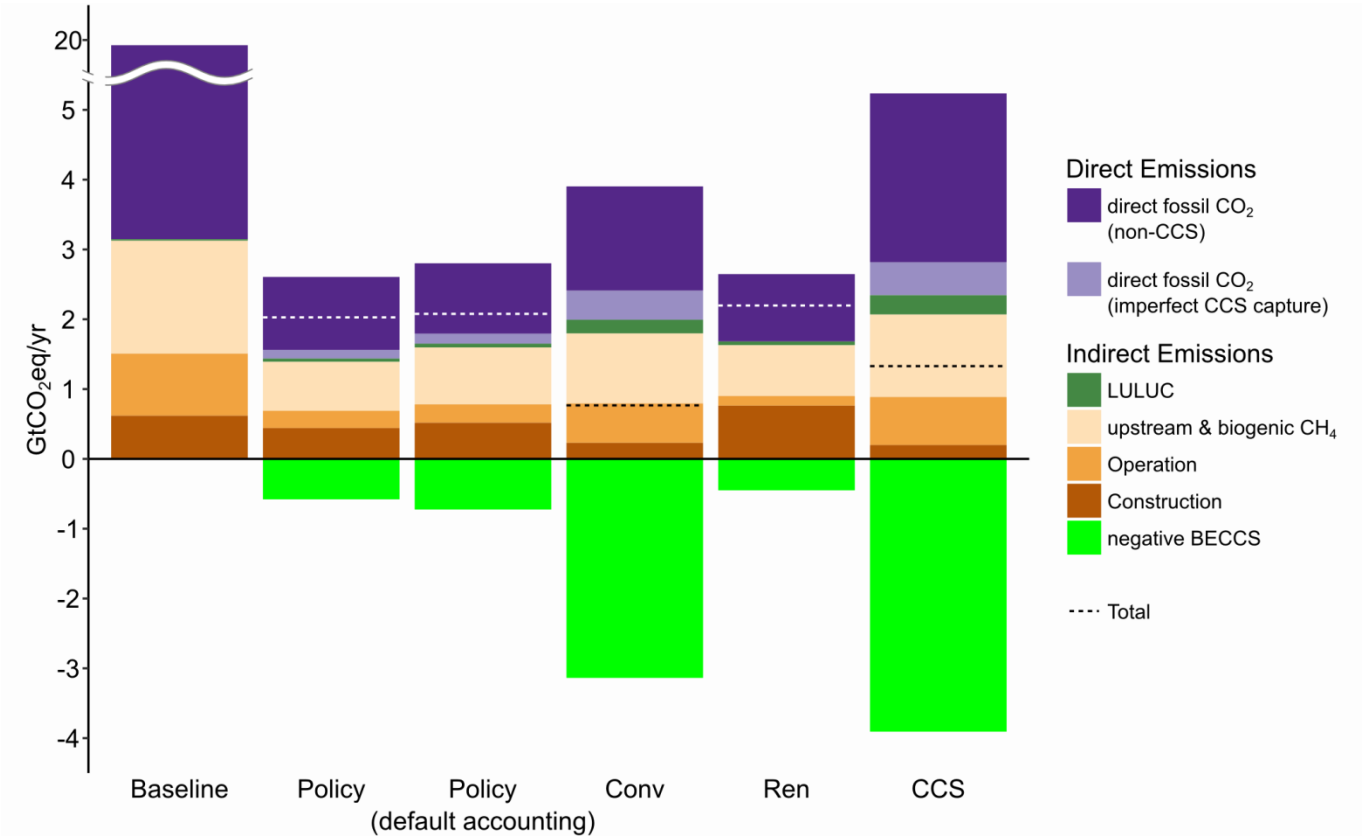
Supplementary Figure 1: ***Scenario overview.*** *Electricity production and capacity additions of the Baseline and four mitigation scenarios. Scenarios "Baseline" and "Policy" account for biogenic CH_4 and EEU CO_2 emissions, "Policy (default accounting)", "Conv", "Ren", and "CCS" do not.*



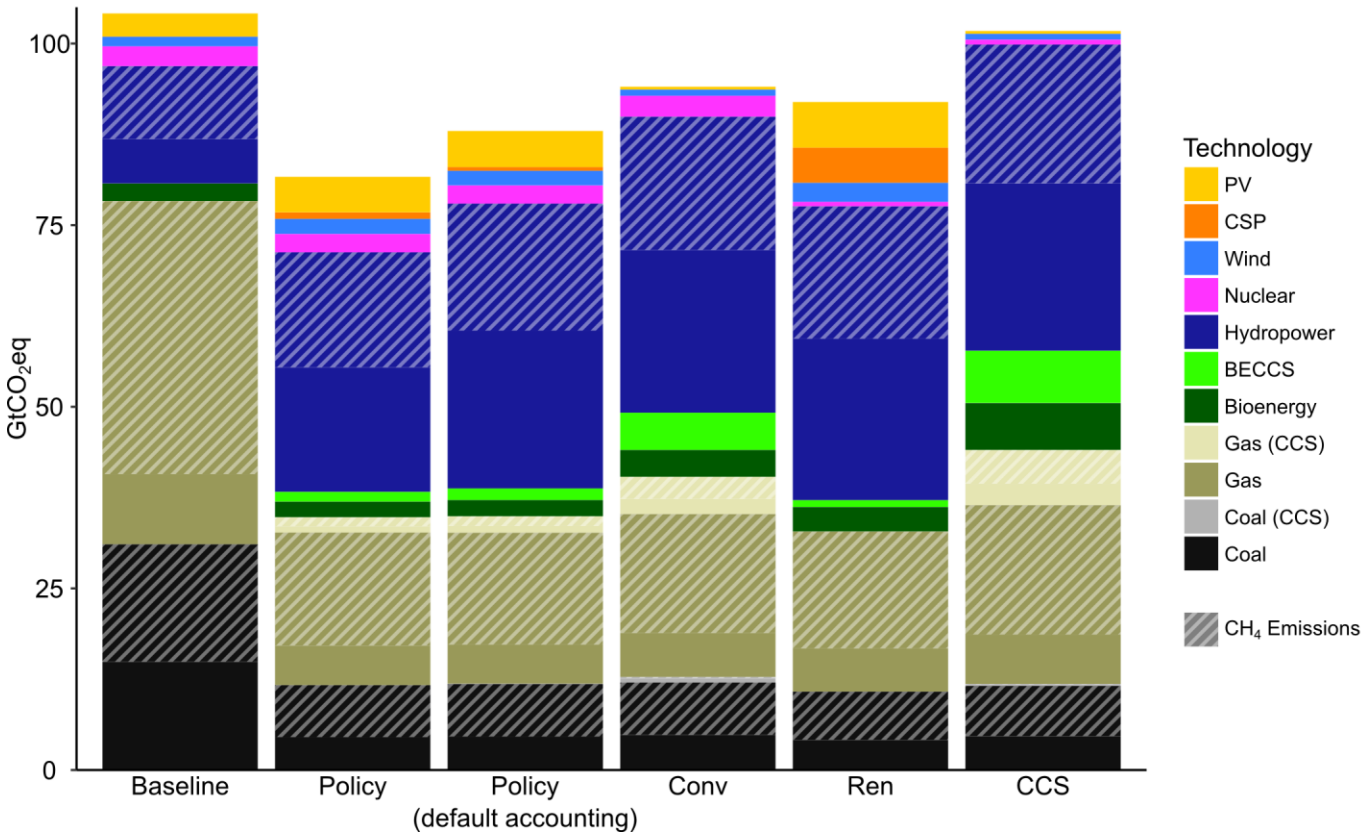
Supplementary Figure 2: ***CO_2-intensities of secondary energy carriers.*** *A robust result of integrated assessment modelling is that the power sector undergoes the fastest and deepest decarbonisation[2,3], as ample low-carbon alternatives to conventional technologies are available. As a result, the CO_2-intensity of electricity drops to nearly zero by 2050 in the climate mitigation scenarios used in this study, while that of solid fuels is halved due to the use of biomass. The CO_2-intensities of gases and liquids remain almost constant on this time horizon.*



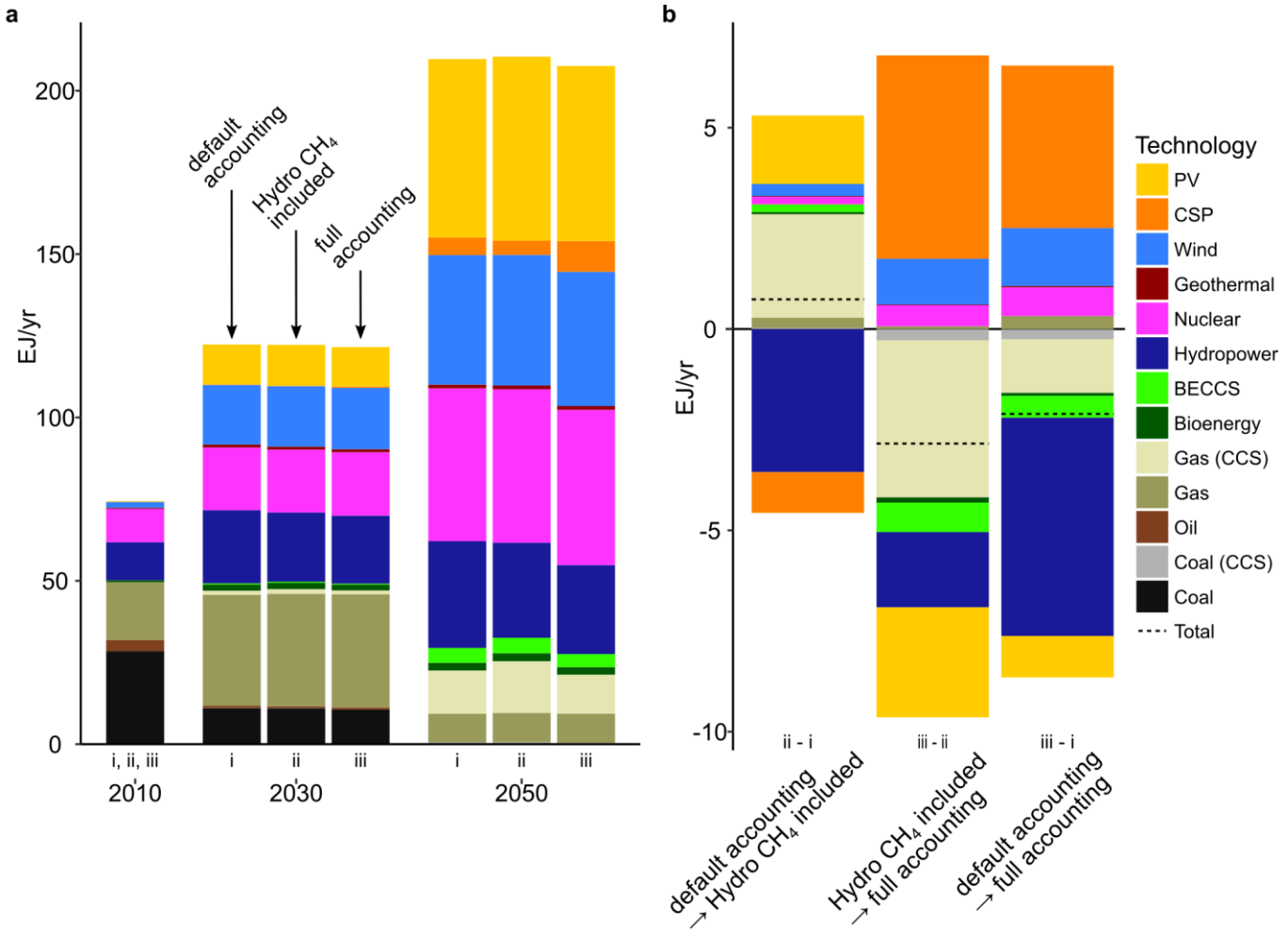
Supplementary Figure 3: ***Temporal patterns of specific GHG emissions.*** *Specific emissions for power sector technologies for 2010 (left bars), 2050 under the Policy scenario (right bars) and a counterfactual scenario with embodied energy use and capacity factors from the Policy scenario, but CO_2-intensities from the Baseline scenario (middle bars). (Since LULUC emissions are computed using 2010 as a base year, no 2010 emissions for bioenergy technologies in 2010 are shown.)*



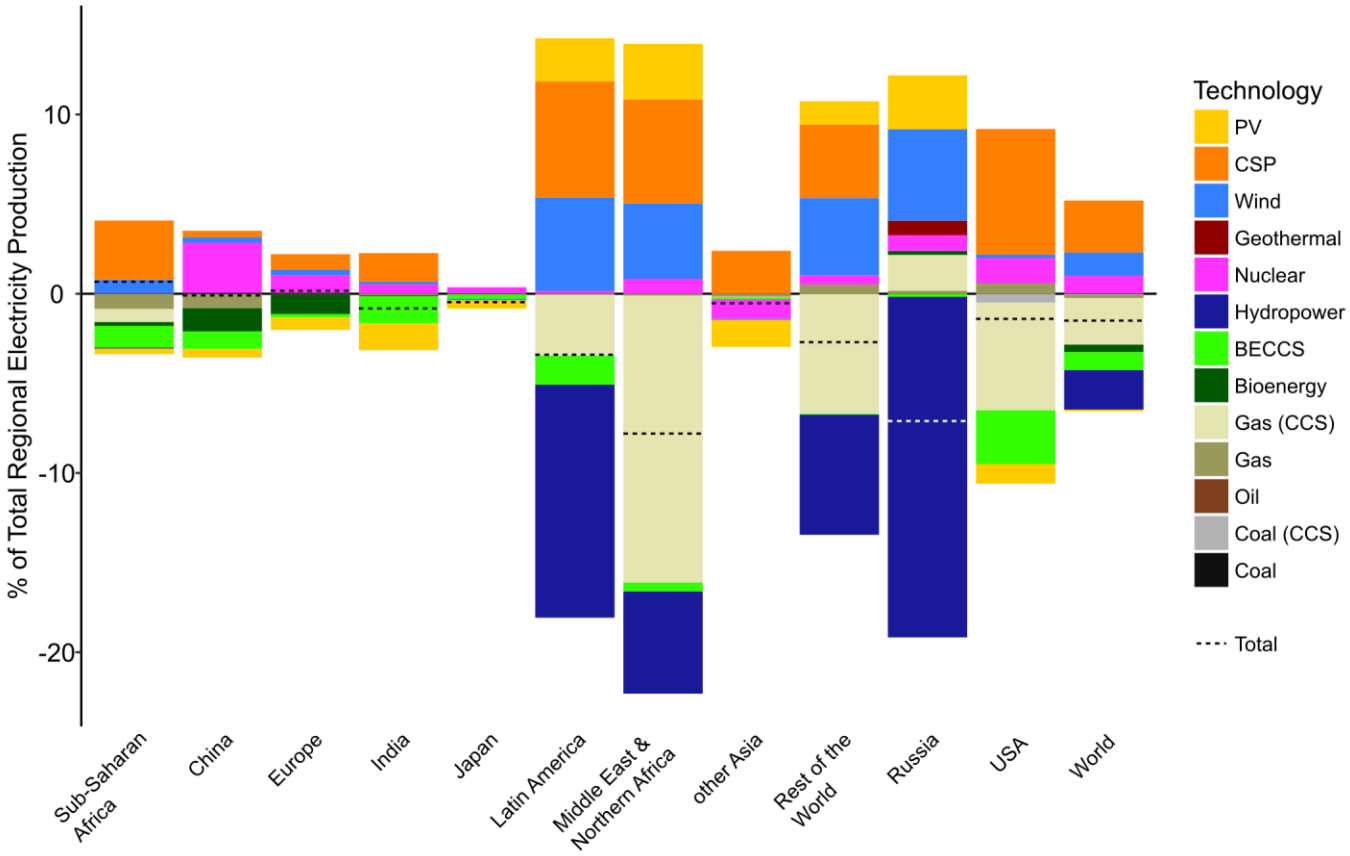
Supplementary Figure 4: ***Total global 2050 emissions.*** *Both direct and indirect emissions from global power production in 2050 for the Baseline and mitigation scenarios. Direct fossil CO_2 (non-CCS) emissions derive from non-CCS fossil plants, while direct fossil CO_2 (imperfect CCS capture) from fossil plants with CCS. See also Supplementary Table 3.*



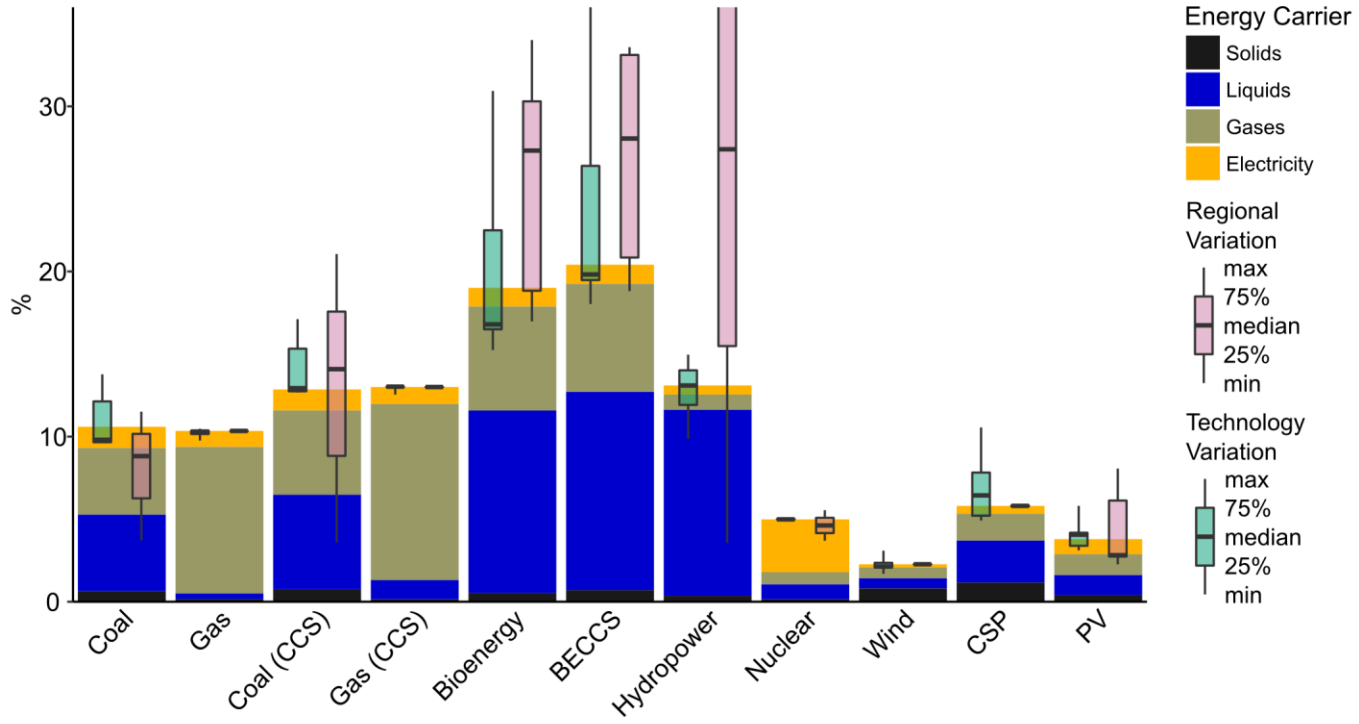
Supplementary Figure 5: ***Cumulated (2010–2050) global indirect emissions for Baseline and mitigation scenarios.*** *Indirect emissions include those from construction, operation (except fuel burning), land use and land-use change (LULUC) and upstream and biogenic CH_4 emissions. See also Supplementary Table 5.*



Supplementary Figure 6: ***Impact on optimal technology choice.*** *Electricity production by technology for three variants of the Policy (2°C-compatible) scenario, accounting only for CH_4 emissions from fossil fuel production and LULUC emissions (default accounting), additionally biogenic CH_4 from hydropower, and all indirect emissions (**a**) and differences between the three variants in 2050 (**b**). Accounting for the effect of indirect emissions so far not considered in IAMs (biogenic CH_4 emissions from hydropower and CO_2 emissions from life-cycle energy use) reduces total electricity production in 2050 slightly. See also Supplementary Table 6.*



Supplementary Figure 7: ***Regional impacts on optimal technology choice.*** *Differences between the full accounting and direct only accounting schemes in 2050 electricity production, as a percentage of total regional electricity production (under direct only accounting). The "World" column corresponds to Figure 4b. See also Supplementary Table 7. While the effects of including indirect GHG emissions in optimal climate change mitigation scenarios is limited on the global scale, regional variations are larger. The overall tendency of decreased production from technologies with comparatively high indirect GHG emissions (gas with CCS, hydropower, bioenergy with and without CCS) which is partly offset by increased production from technologies with comparatively low indirect GHG emissions (CSP, wind, nuclear) is apparent across the regions. Utilisation of PV is reduced to the competition for area with CSP, except for regions that need to offset large reductions in hydropower and natural gas production (Russia, Latin America, and Middle East and Northern Africa). Note that the direct only accounting scheme does not consider abatement options for upstream CH_4 emissions from natural gas production, which reduce these emissions and thus their impact on technology choice. Options for abating biogenic CH_4 emissions from hydropower are not considered at all so far, since insufficient data is available.*



Supplementary Figure 8: ***Embodied energy use of electricity production as a percentage of lifetime electricity production.*** *Global average values by secondary energy carrier (coloured bars) for capacities built in 2050 (identical to Figure 1). Model variations (sample minimum, 25th percentile, median, 75th percentile, and maximum) are shown as boxplots across regions (red) and technology variants (green). See also Supplementary Table 1.*

Supplementary Notes

Supplementary Note 1. Additional Policy Scenarios with Technology Restrictions

To reflect the variety of alternative decarbonisation strategies for the electricity sector, we considered three additional mitigation scenarios which are used throughout the Supplementary Information:

- *Conv*: The combined share of variable renewable electricity (VRE; wind, CSP and PV) is limited to 10%, emphasising the use of conventional technologies and CCS.
- *Ren*: Disallowing the use of CCS in the power sector and investments into nuclear power after 2010 leads to high utilisation of renewable energies.
- *CCS*: Limiting both the use of VREs and nuclear power (as described above) favours the use of fossil fuels with CCS and BECCS.

In order to compare the power-sector response to climate policy on a common basis, the cumulated (net) direct CO_2 emissions of the power sector between 2010 and 2050 were fixed to 240 $GtCO_2$ by iteratively adjusting the initial (2020) level of the global CO_2 tax that applies to all sectors of the economy and grows exponentially with 5% p.a. Since the *Ren* scenario disallows CCS only in the power sector, technologies with CCS in other sectors that have electricity as a co-product may still produce some electricity from biomass using CCS. Co-generation of electricity and other energy carriers was handled by splitting indirect energy use and GHG emissions according to the shares of energy output, on a per-Joule basis. Note that for the additional scenarios biogenic CH_4 emissions from hydropower and CO_2 emissions from embodied energy use (EEU) were not included in the optimisation. To compare them to the *Policy* scenario on an equal footing, we provide an additional scenario that accounts for direct CO_2 emissions upstream CH_4 emissions from coal and gas production and handling, and induced LULUC emissions (CO_2, CH_4, N_2O), named *Policy (default accounting)*.

Supplementary Note 2. Electricity Production and Capacity Additions

Supplementary Figure 1 shows production of electricity and capacity additions through the years 2005–2050 for the *Baseline* and mitigation scenarios. These are the drivers of embodied energy use, either for construction of new capacity (in which end-of-life energy use is subsumed), or for operation and maintenance.

In the *Baseline* scenario, power production increases from 74 EJ/yr in 2010 to 194 EJ/yr in 2050, with conventional technologies (coal and gas without CCS, nuclear and hydropower) providing about three quarters of the total in 2050. Wind and PV contribute more than 10% each, as they become economically competitive even without climate change mitigation policies. Power production is increased to 210 and 215 EJ/yr in the *Policy* and *Ren* scenarios, as electricity is substituted for other energy carriers that are harder to decarbonise [1]. In the *Conv* and *CCS* scenarios, power production in 2050 is 174 and 153 EJ/yr, respectively, as the chief low-carbon technologies (wind, solar PV and CSP) are limited to 10% of total electricity production.

The use of coal is phased out as soon as possible with no new investments in that technology after 2015 in all mitigation scenarios (remaining investments are due to the inertia of capacity investments in the REMIND model). The *Conv* and *CCS* scenarios show some utilisation of coal with CCS. Production from natural gas fills the "gap" between demand and production from renewable sources. It is strongly reduced in the *Policy* and *Ren* scenarios and on the same level as in the baseline or even expanded in the *Conv* and *CCS* scenarios. If gas with CCS is available, it accounts for about two thirds of the production from natural gas. Production from nuclear power is expanded from 10% in the *Baseline* to 22% and 30% in the *Policy* and *Conv* scenarios and is thus an important low-carbon technology, if available. Hydropower is producing 15 EJ/yr in 2050 in the *Baseline* scenario, which is doubled in all policy scenarios. (Biogenic CH_4 emissions are not considered in the optimisation by default.) Variable renewable energies (wind, solar PV and CSP) are the main sources of electricity production in the *Policy*, and *Ren* scenarios, contributing 47 and 75% in 2050, respectively. Bioenergy (with and without CCS) mainly substitutes for limited VREs.

These developments are mirrored in the capacity additions. Investments into non-VRE technologies, which as a rule have high load factors, stay in the range of 0.13–0.33 TW/yr for all except the *Ren* scenarios over the time horizon considered in this study. Investments into VRE technologies reach more than double the value of the baseline scenario (0.49 TW/yr) in the *Policy* (0.85 TW/yr) and *Ren* (1.3 TW/yr) scenarios.

Supplementary Note 3. Temporal Patterns of Indirect Specific Emissions

The embodied energy use of a power sector technology is influenced by three factors: (1) the amount of material and services (e.g. steel for a wind turbine) required for construction and maintenance, (2) the amount of energy the background system needs to provide these services (e.g. energy required per tonne of steel produced) and (3) the energy produced by the technology over its lifetime.

Within our modelling framework, the first two factors (material/service demand and energy intensity of materials/services) are determined within the THEMIS modelling framework. It assumes different electricity mixes for the *Baseline* and mitigation scenarios (following the Baseline and BLUE Map scenarios of ref. [4]) and accounts for probable future improvements in performance parameters for selected industrial processes. It does, however, not account for changes in the characteristics of how transport services are provided. For more details, see [5,6].

The third factor (capacity factor) is characteristic to the energy system technologies as modelled by REMIND. For non-VRE technologies, it is an exogenous, regionally differentiated parameter. For VRE technologies, it is dependent on the amount of energy produced by the individual technologies. Resource potentials for non-biomass renewables (hydropower, wind, solar PV and CSP) are region specific and subdivided by grades, which are characterised by capacity factors. Superior grades have high capacity factors, allowing for high energy production per unit of installed capacity, while inferior grades have lower capacity factors, resulting in lower energy production per unit installed capacity. This grade structure in REMIND contributes to the gradual

expansion of renewable energy production and implies diminishing capacity factors as VRE capacities are expanded.[7]

Specific indirect CO_2 emissions from embodied energy use are furthermore determined by a fourth factor, the CO_2-intensity of the energy carriers, which is endogenous to the REMIND model (see above).

For example, building a wind turbine requires 39 tonnes of cement per MW of capacity. Cement production requires 0.8 GJ of liquid energy carriers per tonne. Over its lifetime, the turbine will produce 178 TJ/MW, resulting in embodied energy use of 0.0487 GJ/kWh_{el} or 0.0175% for this component. Operating a nuclear power plant requires 2.3e-5 GJ/kWh of gases. With a CO_2-intensity of 50.8 $MtCO_2/EJ_{gases}$ the specific indirect emissions from that component are 4.2e-3 gCO_2/kWh in this year.

Supplementary Figure 3 illustrates the contribution of the combined first three factors (material/service demand, energy intensity of materials/services, capacity factors) and the fourth factor (CO_2-intensity) on the specific indirect emissions. To that end, it shows the emissions for 2010 (left bar) and for 2050 under the *Policy* scenario (right bar). The middle bar shows a counter-factual scenario (*Base intensity*), that combines embodied energy use and capacity factors from the *Policy* scenario with CO_2-intensities from the *Baseline* scenario. These specific emissions numbers thus reflect improvements in the technologies themselves as well as the background system, and changing capacity factors (factors 1–3), but not the decrease in CO_2-intensities due to climate mitigation policy (factor 4). The effect of technology improvements (left to middle bars) is most pronounced for PV, which sees its EEU (and thus specific emissions) reduced by two thirds. The other VREs (wind and solar CSP) show only slight reductions due to improved EEU, while nuclear and hydropower show increasing specific emissions, as the capacity factors decrease. This is due to lower grades in the case of hydropower (see above) and competition from renewable energies in the case of nuclear. The effect of reduced CO_2-intensities (middle to right bars) is strongest for nuclear power (a reduction by three fourths), which uses the highest share of electricity and thus benefits most for its decarbonisation.

Supplementary Note 4. Residual Direct and Indirect Power Sector GHG Emissions under Technology Restrictions

While the alternate mitigation scenarios with restricted availability of technologies have markedly different electricity production and investment pathways, they show patterns of total 2050 GHG emissions and cumulated (2010–2050) indirect emissions comparable to the *Policy* scenario, see Supplementary Figures 4 and 5.

The indirect 2050 emissions (Supplementary Figure 4) are on the same order of magnitude (1.4–2.4 $GtCO_2eq$/yr) as in the *Policy* scenario. The different shares of operation and upstream CH_4 emissions and emissions from the construction phase reflect the different utilisation of VRE technologies (wind, CSP and PV), that use only very little energy during operation, as opposed to fossil fuels with CCS, nuclear and bioenergy, that incur most of their indirect emissions during operation. The negative emissions from expanded use of BECCS in the *Conv* and *CCS* scenarios offset higher direct CO_2 emissions due to the limited use of VREs. The net total emissions across the mitigation scenarios are still 90% below that of the *Baseline* scenario, and are lower in 2050 in the *Conv* and *CCS* scenarios than in the *Policy* scenario to compensate for higher emissions earlier on and to adhere to the 240 $GtCO_2$ budget of the scenario design.

The cumulated indirect GHG emissions between 2010 and 2050 (Supplementary Figure 5) differ in details across mitigation scenarios. Indirect emissions from coal and gas without CCS range from 33 to 44 $GtCO_2eq$, while those from hydropower are in the range of 39–42 $GtCO_2eq$ and are markedly lower (33 $GtCO_2eq$) if biogenic CH_4 is accounted for (*Policy* vs. *Policy (default accounting)* scenarios). Emissions from VRE (wind, CSP, PV) are twice as high in the *Ren* scenario (14 $GtCO_2eq$) compared to the *Policy* scenario (7.9 $GtCO_2eq$), while emissions from bioenergy are markedly higher if VREs are not available (8.8–14 $GtCO_2eq$) compared to the *Policy* and Ren

scenarios (3.5–4.3 $GtCO_2eq$). Total cumulated emissions range from 88 to 102 $GtCO_2eq$ across the additional mitigation scenarios and are lowest for the *Policy* scenario which includes them in the optimisation (82 $GtCO_2eq$).

Supplementary Note 5. Effect of Indirect Emissions Price Mark-Ups on Technology Choice

Since many state-of-the-art IAMs already account for upstream CH_4 emissions from fossil fuel production and LULUC emissions, it is of interest to integrated assessment modellers what the impacts of including indirect GHG emissions so far omitted by IAMs are.

Supplementary Figure 6 compares the electricity production in the *Policy* scenario for three accounting schemes: (i) the REMIND *default accounting* that includes upstream CH_4 emissions from coal and gas production as well as LULUC emissions from biomass production, (ii) the *Hydro CH_4 included* scheme which also includes biogenic CH_4 emissions from hydropower, and (iii) the *full accounting* scheme which (as before) also includes price mark-ups for CO_2 emissions from embodied energy use.

Including biogenic CH_4 emissions (ii–i) reduces hydropower production by 11% which is offset by increased production from gas and PV. The overall effect is a slight increase in power production. This and the reduced production from CSP are due to the fact that REMIND optimises power production over its entire time horizon and may change temporal investment patterns to take advantage of reduced costs through technology learning.

Additionally including life-cycle emissions from construction and operation (iii-ii) reduces production from gas, PV, hydropower and bioenergy by 9.3 EJ/yr. Production from CSP, wind and nuclear increases by 6.7 EJ/yr (panel b) and total production falls by 1.4% (panel a). This corresponds to the fact that hydropower, bioenergy and gas technologies have higher specific emissions from construction and operation than nuclear, wind and CSP (cf. Supplementary Table 2). Solar PV production is reduced as it is complemented by production from gas, which allows for balancing of intermittent production.

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